

To Create Pipeline Structures, Store Commands, Maintain Execution Order, Connect Processes, Validate Pipeline Layout, Test Configurations

Code:

```
geya@LenovoL480: ~/OS/Skil × + v
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char input[200];
    char *commands[10];
    int count = 0;

    printf("Enter pipeline: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    /* Split commands using | */
    commands[count] = strtok(input, "|");

    while (commands[count] != NULL && count < 9) {
        count++;
        commands[count] = strtok(NULL, "|");
    }

    printf("\nPipeline structure:\n");

    for (int i = 0; i < count; i++) {
        printf("Command %d: %s\n", i + 1, commands[i]);
    }
}
```

```

printf("\nExecution order:\n");

for (int i = 0; i < count; i++) {
    printf("%s", commands[i]);

    if (i < count - 1)
        printf(" -> ");
}

printf("\n");

/* Validate pipeline */
if (count >= 2)
    printf("Valid pipeline: multiple commands connected.\n");
else
    printf("Invalid pipeline: at least two commands required.\n");

return 0;
}

```

Output:

```

geya@LenovoL480:~/OS/Skill$ gcc task2.c -o task2
geya@LenovoL480:~/OS/Skill$ ./task2
Enter pipeline: ls | grep .c | wc

Pipeline structure:
Command 1: ls
Command 2:  grep .c
Command 3:  wc

Execution order:
ls ->  grep .c ->  wc
Valid pipeline: multiple commands connected.
geya@LenovoL480:~/OS/Skill$ |

```